MIL-E-44258
17 November 1987
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

FOR

ELECTRICAL DISTRIBUTION ASSEMBLIES FOR TENT, EXTENDABLE,

MODULAR, PERSONNEL (TEMPER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of electrical distribution assemblies for use with the TEMPER tent.

1.2 Classification. The electrical distribution assemblies shall be classified as follows (see 6.2):

Type I Commercial connectors	Type II Military standard connectors
Item 1 - Distribution box - (120 V)	Item 1 - Distribution box - 120 V, with MS Cable Leads
Item 2 - Distribution box - (208 V, 3 phase)	Item 2 - Distribution box - 208 V, 3 phase with MS Cable Leads
Item 3 - Convenience outlet assembly - (120 V)	Item 3 - Convenience outlet assembly - (120 V)
Item 4 - Twenty-five foot extension cable for 120 V convenience outlets	Item 4 - Thirty-three foot extension cable for 120 V convenience outlets
Item 5 - Electrical distribution box stand	Item 5 - Electrical distribution box stand
Item 6 - Incandescent light and cable assembly	Item 6 - Incandescent light and cable assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick, Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6150

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic (General Purpose)
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Coraner
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-202 - Test Methods for Electrical and Electrical Component Parts

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-3602 - Cover Plate, Electrical Distribution Box
5-4-3603 - Gasket
5-4-3604 - Cable Assemblies, Receptable Distribution
5-4-3605 - Cable Assemblies, Light Distribution
5-4-3606 - Cable Convenience Outlet Assembly (TEMPER)
5-4-3607 - Cable Convenience Outlet (TEMPER)
5-4-3608 - Cable, Incandescent Light, (TEMPER)
5-4-3622 - Distribution Box,
5-4-3623 - Box, Type A
5-4-3624 - Wiring Diagram, Distribution Box, 120 Volt
5-4-3625 - Distribution Box
5-4-3626 - Box, Type B
5-4-3627 - Cable Assemblies, 3 Phase Receptacle Distribution
5-4-3628 - Wiring Diagram, Distribution Box, 120 Volts

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5-4-3629 - Cable Extension (TEMPER)
5-4-3630 - Electrical Distribution Box Stand
5-4-4139 - Box, 120 Volts
5-4-4140 - Box, 3 Phase, 208 Volts
5-4-4142 - Distribution Box, 120 Volts
5-4-4143 - Cable, Convenience Outlet (TEMPER) (MS)
5-4-4144 - Cable Assemblies, 3 Phase Receptacle Distribution
5-4-4145 - Cable Assemblies, Receptacle Distribution
5-4-4146 - Cable Assemblies, Light Distribution
5-4-4147 - Convenience Outlet Assembly (TEMPER) (MS)
5-4-4148 - Cable Extension (TEMPER) (MS)

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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3.2 Materials and components. Materials and components shall be as specified on the applicable drawings and as specified herein. For components for which the contractor proposes to use an item considered to be equal to the component specified, prior to its use the contractor shall furnish a sample of the component with supporting data to the contracting officer for subsequent evaluation by the responsible military agency. The supporting data required shall prove the functional equivalence and design compatability of the item proposed to be used. Prior to manufacture of the first article, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Construction. Construction shall conform in all respects to the drawings listed in section 2 and as specified herein (see 6.4).

3.4 Performance.

3.4.1 Electrical assemblies.

3.4.1.1 Cables. The electrical cable assemblies, items 3 and 4, shall have a direct current (DC) resistance of less than 1 ohm when tested for electrical continuity as specified in 4.4.4. In addition, insulation resistance shall be greater than 100 megohms when tested as specified in 4.4.4.

3.4.1.2 Distribution boxes. The electrical distribution boxes, items 1 and 2, shall have a DC resistance of less than 1 ohm when tested for electrical continuity as specified in 4.4.4.

3.5 Workmanship. The electrical distribution components assemblies shall conform to the quality of the product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the applicable defects specified in 4.4.2 and 4.4.3 and tested for the applicable characteristics specified in table II. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for applicable defects listed in table I. The lot size shall be expressed in units of electrical distribution assemblies of one item number only. The sample unit shall be one electrical distribution assembly. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Distribution boxes	Broken	X	
	Malformed	X	
	Switches fail to operate	X	
	Corroded areas		X
	Burrs or sharp edges	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Distribution box stands	Broken	X	
	Malformed	X	
	Corroded areas		X
	Burrs or sharp edges	X	
	Welds improperly located	X	
	Welds containing any pits, protrusions, or voids		X
Cable wires	Evidence of burns, cracking, abrasions, cuts, or pinch marks	X	
Soldered connections	Rough or incomplete	X	
	Poorly formed		X
	Lack of adherence	X	
	Dull or dirty appearance		X
	Presence of protrusions, pits, or voids		X
Cleanliness	Grease or oil stains on components, excess flux or solder		X
Marking and labels	Omitted, incorrect, illegible, or misplaced		X

4.4.3 End item dimensional examination. The end items shall be examined for compliance with dimensions that are identified on the drawings. Any dimension not with applicable tolerance shall constitute a defect. The lot size shall be expressed in units of electrical distribution assemblies of one item number only. The sample unit shall be one electrical distribution assembly. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 End item testing. Testing of the electrical distribution assemblies shall be performed in accordance with table II. The lot size shall be expressed in units of electrical distribution assemblies of one item number only. The sample size shall be the number of assemblies indicated by inspection level S-1. Any test failure shall be cause for rejection of the lot.

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TABLE II. End item tests

Characteristics	Requirement paragraph ref.	Test method
Electrical continuity, items 3 and 4	3.4.1.1	4.5.1
Insulation resistance, items 3 and 4	3.4.1.1	4.5.2
Distribution box continuity, items 1 and 2	3.4.1.2	4.5.3

4.4.4 Packaging examination. The fully packaged end items shall be examined for applicable defects listed in table III. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing; any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Contents	Number per container is more or less than required.

4.5 Methods of inspection.

4.5.1 Electrical continuity for cable assemblies. Each conductor in the electrical cable assembly shall be tested in accordance with Method 303 of MIL-STD-202. The test current shall not exceed 15 milliamperes. Measurement shall be made between connector pins which are mutually common to each other electrically. The continuity for each pin shall be checked for compliance with the continuity requirement of 3.4.1.1. Failure to comply shall be considered a test failure.

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4.5.2 Insulation resistance for cable assemblies. The cable assembly shall be tested in accordance with Method 302 of MIL-STD-202. The test potential shall be 500 volts, test condition B. Insulation resistance measurement shall be made between the mutually insulated points and between insulated points and connector shell or ground. Each resistance measurement shall be checked for compliance with the continuity requirement of 3.4.1.1. Failure to comply shall be considered a test failure.

4.5.3 Electrical continuity of distribution boxes. Each circuit in the electrical distribution box shall be tested in accordance with Method 303 of MIL-STD-202. The test current shall not exceed 15 milliamperes. Measurement shall be made between each terminal pin of the circuit which is electrically common to each other terminal pin. The continuity for each pin shall be checked for compliance with the continuity requirements of 3.4.1.2. Failure to comply shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as applicable (see 6.2).

5.1.1 Level A.

5.1.1.1 Items 1, 2, 3, 4 and 6. All cables shall be individually coiled into loops of approximately 12 inches in diameter and secured with a plastic coated metal twist tie. Each item shall then be inserted in a close-fitting clear polyethylene film bag having a minimum thickness of 0.004 inches conforming to type II, style 2 of PPP-B-26. Each bagged item shall be unit packed in a snug-fitting fiberboard box conforming to style FOL, (variety SW), class domestic, grade 275 of PPP-B-636. Cushioning shall be used as necessary to prevent movement or damage during transit. Closure shall be in accordance with PPP-B-636.

5.1.1.2 Item 5. The box stand shall be unit packed in a snug-fitting fiberboard box conforming to style FPF, grade V3c of PPP-B-636. Closure shall be in accordance with PPP-B-636.

5.1.2 Commercial. Each item shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Items, preserved as specified in 5.1, shall be packed in the quantities specified in table IV in a cleated plywood container conforming to overseas type, style A or B, type 2 load of PPP-B-601, or a nailed wood shipping container conforming to class 2, style 2, 2-1/2, 4, 4-1/2, or 5 of PPP-B-621. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

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TABLE IV. Packing quantities

Item	Quantity per shipping container
1	1
2	1
3	12
4	12
5	12
6	12

5.2.2 Level B packing.

5.2.2.1 Items 1, 2, 3, 4, and 6. Items 1, 2, 3, 4, and 6, preserved as specified in 5.1, shall be packed in the quantities specified in table IV in a snug-fitting fiberboard shipping container conforming to FOL, grade V11c or V13c of PPP-B-636. Each shipping container shall be closed and reinforced as specified in the appendix of PPP-B-636.

5.2.2.2 Item 5. Item 5, unit packed as specified in 5.1.1.2, shall not require additional packing.

5.2.3 Commercial packing. Items, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.2.3 Marking. In addition to any special marking by the contract or purchase order, unit packs and shipping containers shall be marked in acccordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The electrical distribution assemblies are designed to supply adequate electrical power to lighting fixtures and to the electrical outlets of the TEMPER tent.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Types and items required (see 1.2).
c. When a first article sample is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable level of preservation and packing (see 5.1 and 5.2).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangments for selection, inspection, and approval of the first article.

6.4 Drawings. The drawings necessary for construction of individual types and items are as follows:

Type I	Item 1	Item 2	Item 3
	5-4-3622	5-4-3625	5-4-3606
	5-4-3602	5-4-3602	5-4-3607
	5-4-3603	5-4-3603	Item 4
	5-4-3604	5-4-3605	5-4-3629
	5-4-3605	5-4-3626	Item 5
	5-4-3623	5-4-3627	5-4-3630
	5-4-3624	5-4-3628	Item 6
			5-4-3608

Type II	Item 1	Item 2	Item 3
	5-4-4141	5-4-4142	5-4-4147
	5-4-3602	5-4-3602	5-4-4143
	5-4-3603	5-4-3603	Item 4
	5-4-3624	5-4-3628	5-4-4148
	5-4-4139	5-4-4140	Item 5
	5-4-4145	5-4-4144	5-4-3630
	5-4-4146	5-4-4146	Item 6
			5-4-3608

6.5 Supersession data. This document includes the requirements of LP/P. DES 3-83A, dated 13 February 1985.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document.

6.7 Subject term (key word) listing.

Electrical distribution
Power supply, Temper test
TEMPER

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Custodians:

Army - GL
Air Force - 99

Review activities:

Air Force - 82
DLA - GS

Preparing activity:

Army - GL

Project No. 6150-0173

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-44258	Electrical Distribution Assemblies for Tent, Extendable, Modular, Personnel (TEMPER)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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